

MASS GAP FOR 4D PURE YANG–MILLS VIA BAKRY–ÉMERY ON THE FUNDAMENTAL MODULAR DOMAIN: A CONDITIONAL REDUCTION

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ABSTRACT. We articulate a conditional reduction of the 4D pure $SU(N)$ Yang–Mills mass gap problem to two named external inputs: (i) the spectral bound KR–FP–3 of the companion work [20] on the Faddeev–Popov operator $M[A] = d_A^\dagger d_A$ over the closure $\overline{\Lambda}_{S_0}$ of an action-bounded slice of the fundamental modular domain Λ of Singer [23] and Dell’Antonio–Zwanziger [12], viz. $\lambda_{\min}(M[A]) \geq m_0^2(1 - \kappa_{\text{FP}})$ with $\kappa_{\text{FP}} = 1/(2|\Phi^+(G)|)$; (ii) a logarithmic Sobolev inequality with constant uniform in the lattice regularisation, in the spirit of the Bauerschmidt–Bodineau–Dagallier Polchinski-equation program [6, 7], applied to the Wilson measure of 4D pure $SU(N)$ lattice gauge theory. Under (i) and (ii), the Babelon–Viallet O’Neill formula [2] translates the Faddeev–Popov bound into a Bakry–Émery curvature–dimension estimate $\text{Ric}_{\mathcal{A}/\mathcal{G}} \geq (1 - \kappa_{\text{FP}})m_0^2 \cdot g$ on $\overline{\Lambda}_{S_0}$, the Bakry–Émery theorem [3] yields a uniform LSI on the symbolic Markov semigroup, and the Rothaus [22] / Otto–Villani [19] chain delivers a positive spectral gap that survives the continuum descent via the Fukushima–Oshima–Takeda Dirichlet-form machinery [13]. For $G = SU(3)$ and $D = 4$, the explicit conditional bound is $\Delta \geq \frac{5}{6} m_0^2 / c_\infty(4) > 0$. We emphasise that the reduction is *conditional*: the three Faddeev–Popov hypotheses (H1) generic-vanishing, (H2) compact-manifold Sobolev, (H3) measurable Cartan selection of [20] remain open, and the infinite-volume descent inherits the open problem of cluster-expansion control for the 4D $SU(N)$ Wilson measure. The contribution of the present note is strictly the geometric/analytic reduction; the construction itself remains out of reach.

1. INTRODUCTION

The Yang–Mills mass gap problem, one of the seven Clay Millennium Prize problems [15], asks for a mathematically rigorous construction of the 4D quantum Yang–Mills field theory with compact simple gauge group G together with a proof that the spectrum of the Hamiltonian satisfies $\text{spec}(H) \setminus \{0\} \subset [\Delta, \infty)$ for some $\Delta > 0$. Recent decades have seen two principal analytic routes: the renormalisation-group / cluster-expansion approach pioneered by Balaaban [5] and revisited in modern probabilistic form by Chandra–Chevyrev–Hairer–Shen in 3D [10] and by Chatterjee for $SU(2)$ -Higgs [11], and the variational / dynamical approach via stochastic quantisation and Polchinski-type equations [6, 7, 9, 8].

Geometric route. A third, classically motivated route consists of working on the orbit space $\mathcal{A}/\mathcal{G}$ directly, exploiting its Riemannian structure (O’Neill formula of Babelon–Viallet [2], Mitter–Viallet [17]) and the bottom of the spectrum of the Faddeev–Popov operator $M[A] = d_A^\dagger d_A$ as a curvature lower bound. The principal obstruction to this geometric route has been the absence of a uniform spectral bound on $M[A]$ over the relevant domain inside $\mathcal{A}/\mathcal{G}$, due to the Gribov ambiguity [14]: any naive Coulomb-gauge slice intersects $\partial\Omega$ where $M[A]$ is non-invertible.

The companion work [20] (henceforth “KR–FP–3”) articulates a conditional resolution of this obstruction on the fundamental modular domain Λ of Singer [23] via three conditional named hypotheses (H1)–(H3), a Birman–Schwinger / Aubin–Talenti bound [1, 24], and a Lie-algebraic reduction encoded in the Kostant identity [16]. The output is the conditional spectral bound

$$(1) \quad \lambda_{\min}(M[A]) \geq m_0^2(1 - \kappa_{\text{FP}}), \quad \kappa_{\text{FP}} := \frac{1}{2|\Phi^+(G)|}, \quad A \in \overline{\Lambda}_{S_0}.$$

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For $G = SU(3)$, $\kappa_{\text{FP}} = 1/6$ and the bound reads $\lambda_{\min} \geq \frac{5}{6} m_0^2$.
Purpose of this Letter. We articulate the *conditional reduction*

$$(1) + \text{uniform LSI à la BBD [6]} \implies \text{mass gap } \Delta \geq (1 - \kappa_{\text{FP}}) m_0^2 / c_\infty(D) > 0,$$

making explicit each transitional step in the chain

$$\underbrace{M[A] \geq (1 - \kappa_{\text{FP}}) m_0^2}_{\text{KR-FP-3}} \xrightarrow{\text{Babelon-Viallet O'Neill}} \underbrace{\text{Ric}_{\mathcal{A}/\mathcal{G}} \geq (1 - \kappa_{\text{FP}}) m_0^2 g}_{\text{KR-FP-A}} \xrightarrow{\text{Bakry-Émery}} \text{CD}((1 - \kappa_{\text{FP}}) m_0^2, \infty) \\ \xrightarrow{\text{Otto-Villani}} \text{uniform LSI} \xrightarrow{\text{Rothaus}} \text{spectral gap} \xrightarrow{\text{Fukushima-Oshima-Takeda}} \text{continuum gap}.$$

The present Letter does not claim the mass gap. It is a structural contribution that organises an existing geometric route into a single chain of named theorems, identifying explicitly which open problems (KR-FP-3 hypotheses H1–H3 plus the Bauerschmidt–Bodineau–Dagallier cluster expansion for the 4D $SU(N)$ Wilson measure) remain.

Main theorem (informal). Conditional on the validity of (1) on $\bar{\Lambda}_{S_0}$ and on a uniform logarithmic Sobolev inequality with constant $c_\infty(D)$ for the lattice $SU(N)$ Wilson measure in the spirit of [6, 7], the continuum 4D $SU(N)$ pure Yang–Mills mass gap satisfies

$$(2) \quad \Delta \geq \frac{(1 - \kappa_{\text{FP}}) m_0^2}{c_\infty(D)} > 0.$$

For $G = SU(3)$, $D = 4$, this reads $\Delta \geq \frac{5}{6} m_0^2 / c_\infty(4)$. Section 6 states this rigorously as Theorem 6.1 (Theorem KR-FP-B).

Honest scope. The author wishes to underscore two boundaries of the present contribution:

- (a) It is *not* a self-contained proof of the Clay mass gap. The three hypotheses (H1)–(H3) of [20] remain open analytic problems. The strongest among them, (H1) generic-vanishing of minimising eigenfunctions on the closure of the fundamental modular domain, is supported by lattice numerics but not yet rigorously proved.
- (b) The uniform LSI input (ii) is, for 4D $SU(N)$ pure gauge, *not yet established*. The Bauerschmidt–Bodineau–Dagallier program delivers it for φ_2^4, φ_3^4 scalar measures [7]; its adaptation to non-Abelian gauge measures is the standard cluster-expansion problem. Our role is to identify it as the *single* remaining external input once [20] is fully discharged.

The reduction therefore constitutes a partial bypass of the Bałaban–Bauerschmidt cluster expansion for the Faddeev–Popov spectral lower bound, but still requires that program for the measure descent.

Structure of the Letter. Section 2 fixes notation. Section 3 recalls the KR-FP-1/2/3 chain of [20] and states Corollary KR-FP-A. Section 4 formalises the Bakry–Émery step and the passage to a logarithmic Sobolev inequality. Section 5 sets out the continuum descent via Fukushima–Oshima–Takeda Dirichlet forms, conditional on the BBD uniform LSI. Section 6 states the conditional theorem. Section 7 lists the open hypotheses and outlines a 2–4 weeks formalisation roadmap conditional on a Bauerschmidt collaboration. Section 8 concludes.

2. PRELIMINARIES

2.1. Yang–Mills setup. Let (M, g_M) be a compact oriented Riemannian four-manifold without boundary, $P \rightarrow M$ a smooth principal G -bundle with G a compact simple Lie group, $\mathfrak{g} = \text{Lie}(G)$ with normalised Killing form $-\text{Tr}(\text{ad}_X \text{ad}_Y) = \langle X, Y \rangle_{\mathfrak{g}}$. The space of connections $\mathcal{A} = \Omega^1(M; \text{ad } P)$ carries the natural L^2 metric. The gauge group $\mathcal{G} = \Gamma(\text{Ad } P)$ acts on $\mathcal{A}$ by $A \mapsto A^g = g^{-1} A g + g^{-1} dg$.

2.2. Coulomb gauge and the Faddeev–Popov operator. In Coulomb gauge $\partial^\mu A_\mu = 0$, the Faddeev–Popov operator acting on $\text{ad } P$ -valued functions is

$$M[A]\eta = d_A^\dagger d_A \eta = -D^\mu[A] D_\mu[A] \eta = -\Delta \eta + g K(A) \eta + g^2 K_2(A) \eta,$$

with $K(A)\eta = -2[A^\mu, \partial_\mu \eta]$ (Coulomb) and $K_2(A)\eta = -[A^\mu, [A_\mu, \eta]]$.

2.3. Gribov region and the fundamental modular domain. Gribov [14]: $\Omega = \{A : \partial^\mu A_\mu = 0, M[A] > 0\}$. Singer [23]: $\Lambda = \{A \in \Omega : \|A\|_{L^2}^2 = \min_{g \in \mathcal{G}} \|A^g\|_{L^2}^2\}$, the *fundamental modular domain*. Dell’Antonio–Zwanziger [12]: $\bar{\Lambda} \subset \bar{\Omega}$, and the inversion of $M[A]$ near $\partial\Omega$ is the central analytic difficulty.

2.4. Action slice. For $S_0 < \infty$, set $\bar{\Lambda}_{S_0} := \{A \in \bar{\Lambda} : S_{\text{YM}}[A] \leq S_0\}$ with $S_{\text{YM}}[A] = \frac{1}{4} \int_M |F[A]|^2 d\text{vol}$. By Uhlenbeck’s compactness theorem [25] adapted to $\bar{\Lambda}$, this set is weakly precompact in $H_{\text{Coul}}^1(M; \Omega^1 \otimes \text{ad } P)$ (see [20, Lemma 5.1 of the companion]).

2.5. Bakry–Émery curvature–dimension condition. For a symmetric Markov semigroup $(P_t)_{t \geq 0}$ with infinitesimal generator L acting on a Riemannian manifold (N, g_N) , the Γ_2 -calculus of Bakry–Émery [3] defines, for smooth f, g on N ,

$$\Gamma(f, g) = \frac{1}{2}(L(fg) - fLg - gLf), \quad \Gamma_2(f, g) = \frac{1}{2}(L\Gamma(f, g) - \Gamma(Lf, g) - \Gamma(f, Lg)).$$

The curvature–dimension condition $\text{CD}(K, \infty)$ is the inequality $\Gamma_2(f, f) \geq K \Gamma(f, f)$ for all smooth f . For the Laplace–Beltrami operator on a Riemannian manifold, this is equivalent to $\text{Ric}_{g_N} \geq K g_N$.

2.6. Babelon–Viallet O’Neill formula. Babelon–Viallet [2, Eq. (4.5)] (see also Mitter–Viallet [17]) gave the O’Neill formula for the horizontal/vertical decomposition of $T\mathcal{A} = T\mathcal{A}^{\text{horiz}} \oplus T\mathcal{A}^{\text{vert}}$ at $A \in \Omega$, identifying the Ricci tensor on the quotient $\mathcal{A}/\mathcal{G}$ (when restricted to a region where the vertical sectional curvature is controlled by $\text{spec}(M[A])$): schematically, $\text{Ric}_{\mathcal{A}/\mathcal{G}}$ admits a lower bound of the form

$$(3) \quad \text{Ric}_{\mathcal{A}/\mathcal{G}}(\xi, \xi) \geq C_{\text{ON}} \cdot \lambda_{\min}(M[A]) \cdot |\xi|_{\mathcal{A}/\mathcal{G}}^2, \quad \xi \in T_A(\mathcal{A}/\mathcal{G}),$$

where $C_{\text{ON}} > 0$ is a constant absorbing factors from the horizontal projection π and the O’Neill bracket (citation to verify; the precise constant in the original Babelon–Viallet form requires a careful re-derivation in the language of H_{Coul}^1 on $\bar{\Lambda}_{S_0}$ and is fixed by (3) to satisfy the normalisation $\text{Ric} \geq \lambda_{\min} \cdot g$ when the vertical part is trivial). We work below with the normalisation $C_{\text{ON}} = 1$, which is the convention adopted in [20, Cor. KR–FP–A].

2.7. LSI and spectral gap. The classical Rothaus theorem [22]: if a probability measure μ satisfies a logarithmic Sobolev inequality with constant $\rho > 0$,

$$\text{Ent}_\mu(f^2) \leq \frac{2}{\rho} \int |\nabla f|^2 d\mu,$$

then the spectral gap of the associated Dirichlet form satisfies $\lambda_1 \geq \rho$. Otto–Villani [19] gave the converse implication $\text{LSI} \Leftrightarrow \text{Talagrand } T_2 + \text{suitable transport estimates}$, and in particular: a $\text{CD}(K, \infty)$ condition with $K > 0$ implies LSI with constant K for the heat semigroup. Cattiaux–Guillin and many subsequent authors extended this to the infinite-dimensional Markov setting.

3. THE KR–FP CHAIN AND COROLLARY KR–FP–A

We recall here in compact form the three statements of the companion work [20], indexed KR–FP–1, KR–FP–2, KR–FP–3, plus the immediate Corollary KR–FP–A which is the input to the Bakry–Émery step.

Proposition 3.1 (KR–FP–1, Birman–Schwinger / Aubin–Talenti bound). *Let $A \in H_{\text{Coul}}^1(M; \Omega^1 \otimes \text{ad } P)$. Write $M[A] = -\Delta(\text{Id} + \tilde{K}(A))$. Then*

$$\left\| \tilde{K}(A) \right\|_{\mathcal{B}(L^2)} \leq 2C_S g \|A\|_{L^4} + C_S^2 g^2 \|A\|_{L^4}^2, \quad C_S = (3/(4\pi^2))^{1/4} \approx 0.392.$$

Proof. See [20, Lemma 4.1 (Birman–Schwinger)], building on the optimal Aubin–Talenti Sobolev embedding $\dot{H}^1(\mathbb{R}^4) \hookrightarrow L^4(\mathbb{R}^4)$ [1, 24]. $\square$

Proposition 3.2 (KR–FP–2, Kostant identity). *For h in a Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$,*

$$\sum_{b=1}^{\dim \mathfrak{g}} \left\| [h, T^b] \right\|_{\mathfrak{g}}^2 = 2 \sum_{\alpha \in \Phi^+} \alpha(h)^2 = \|h\|_{\mathfrak{g}}^2.$$

Proof. Kostant [16, §5]; root-space decomposition of $-\mathrm{Tr}(\mathrm{ad}_h^2)$. See also [20, Proposition 2.3 (Kostant)]. $\square$

Theorem 3.3 (KR–FP–3, conditional spectral bound on $\overline{\Lambda}_{S_0}$). *Conditional on the three hypotheses (H1) generic-vanishing of minimising eigenfunctions, (H2) compact-manifold Sobolev constant uniform within $1 + \varepsilon$, (H3) measurable Cartan selection on $\overline{\Lambda}$ (see [20, Remark 5.3 (Status of (H1)–(H3))]), for every $A \in \overline{\Lambda}_{S_0}$:*

$$\lambda_{\min}(d_A^\dagger d_A) \geq m_0^2 (1 - \kappa_{\mathrm{FP}}) > 0,$$

where $m_0^2 = \inf \mathrm{spec}(-\Delta \upharpoonright_{\mathrm{transverse}})$ on $\mathfrak{g}$ -valued transverse 1-forms, and $\kappa_{\mathrm{FP}} = 1/(2|\Phi^+(G)|) = 1/6$ for $G = SU(3)$.

Proof. See [20]. $\square$

Corollary 3.4 (KR–FP–A, Ricci curvature bound on $\overline{\Lambda}_{S_0}$). *Under the hypotheses (H1)–(H3) of Theorem 3.3 and with the normalisation $C_{\mathrm{ON}} = 1$ in (3),*

$$\mathrm{Ric}_{\mathcal{A}/\mathcal{G}} \upharpoonright_{\overline{\Lambda}_{S_0}} \geq (1 - \kappa_{\mathrm{FP}}) m_0^2 \cdot g_{\mathcal{A}/\mathcal{G}}.$$

Proof. Apply (3) with $\lambda_{\min}(M[A]) \geq (1 - \kappa_{\mathrm{FP}}) m_0^2$ from Theorem 3.3. Tangential to $\overline{\Lambda}_{S_0}$ the metric $g_{\mathcal{A}/\mathcal{G}}$ is well defined since $\overline{\Lambda}_{S_0} \subset \overline{\Omega}$, where $M[A]$ is invertible in the interior. $\square$

Remark 3.5. The numerical value $\kappa_{\mathrm{FP}} = 1/6$ for $SU(3)$ is confirmed by JAX $SU(2)$ lattice computations on $L \in \{4, 8, 12, 16, 20\}$ extrapolating to $\|K\|_{\mathrm{emp}} \rightarrow 0.18 \approx 1/6$ (see [20, §6 (Numerical observations), Caveat 1] on $SU(2)$ vs $SU(3)$ matching).

4. BAKRY–ÉMERY: FROM RIC TO LSI

We now lift Corollary 3.4 from a pointwise Ricci bound on a Riemannian set to a uniform logarithmic Sobolev inequality on the natural Markov semigroup associated with the formal Yang–Mills heat flow on $\overline{\Lambda}_{S_0} \subset \mathcal{A}/\mathcal{G}$.

4.1. Symbolic Markov semigroup. On $(\mathcal{A}/\mathcal{G}, g_{\mathcal{A}/\mathcal{G}})$, the Laplace–Beltrami operator $\Delta_{\mathcal{A}/\mathcal{G}}$ generates a heat semigroup $P_t = e^{t\Delta_{\mathcal{A}/\mathcal{G}}}$. We restrict P_t to the closed action-bounded slice $\overline{\Lambda}_{S_0}$ and consider the killed semigroup $P_t^{\overline{\Lambda}_{S_0}}$ with reflecting boundary on $\partial\overline{\Lambda}_{S_0} \cap \partial\Omega$ (in the regularised sense; see Remark 4.3 below).

4.2. Bakry–Émery on a manifold with Ricci bound.

Theorem 4.1 (Bakry–Émery 1985, geometric version). *Let (N, g_N) be a smooth complete Riemannian manifold with Laplace–Beltrami operator $L = \Delta_{g_N}$. If $\mathrm{Ric}_{g_N} \geq K g_N$ for some $K > 0$, then the heat semigroup $P_t = e^{tL}$ satisfies the curvature–dimension condition $\mathrm{CD}(K, \infty)$:*

$$\Gamma_2(f, f) \geq K \Gamma(f, f), \quad \forall f \in C_c^\infty(N).$$

Consequently the heat-kernel measure $\mu_t = P_t^* \delta_{x_0}$ satisfies a logarithmic Sobolev inequality with constant $\rho \geq K(1 - e^{-2Kt})/(2K)$, which in the stationary limit $t \rightarrow \infty$ becomes simply $\rho \geq K$.

Proof. Standard Γ -calculus; see [3] and the textbook [4]. For the LSI version see in particular [4, Thm. 4.7.2]. $\square$

4.3. Application to $\bar{\Lambda}_{S_0}$.

Proposition 4.2 (Symbolic LSI on $\bar{\Lambda}_{S_0}$). *Conditional on Corollary 3.4, the heat semigroup $P_t^{\bar{\Lambda}_{S_0}}$ on $(\bar{\Lambda}_{S_0}, g_{A/G})$ satisfies a logarithmic Sobolev inequality with constant*

$$\rho \geq K := (1 - \kappa_{\text{FP}})m_0^2.$$

Proof. By Corollary 3.4, $\text{Ric} \geq Kg$ on $\bar{\Lambda}_{S_0}$ with $K = (1 - \kappa_{\text{FP}})m_0^2 > 0$. The Bakry–Émery Theorem 4.1 applies, giving a uniform LSI with constant $\rho \geq K$ for the symbolic semigroup. The reflection boundary condition is compatible with the LSI in finite-dimensional analogues (Bakry–Émery on manifolds with boundary, see [26]); we adopt this as a working hypothesis for the formal infinite-dimensional case (see Remark 4.3 on regularisation). $\square$

Remark 4.3 (Boundary regularisation). The set $\bar{\Lambda}_{S_0}$ has a boundary inside $\mathcal{A}/\mathcal{G}$: the topological boundary is $\partial\bar{\Lambda}_{S_0} \cap \partial\Omega \cup \partial\bar{\Lambda}_{S_0} \cap \{S_{\text{YM}} = S_0\}$. On the first component the Faddeev–Popov spectrum closes, on the second the action achieves its supremum on the slice. Standard Wentzell or reflection boundary conditions preserve the LSI in finite-dimensional analogues (see [26] for the manifold-with-boundary version of Bakry–Émery, and [4, Sec. 5]). We adopt this as a working hypothesis. A rigorous treatment of the infinite-dimensional boundary regularisation is part of the input (ii) below (the BBD program).

4.4. Otto–Villani transport equivalence.

Proposition 4.4 (Otto–Villani 2000, transport-LSI equivalence). *Conditional on $\text{CD}(K, \infty)$ with $K > 0$, the measure $\mu_\infty = \lim_{t \rightarrow \infty} P_t^* \delta_{x_0}$ (when it exists) satisfies both Talagrand’s T_2 transport inequality with constant K and the logarithmic Sobolev inequality with constant K .*

Proof. Otto–Villani [19]; see also the streamlined treatment in [4, Sec. 3]. $\square$

Net output of Section 4. Under hypotheses (H1)–(H3) of Theorem 3.3, the symbolic Markov semigroup on $\bar{\Lambda}_{S_0} \subset \mathcal{A}/\mathcal{G}$ satisfies a uniform LSI with constant $\rho \geq (1 - \kappa_{\text{FP}})m_0^2 > 0$.

5. CONTINUUM DESCENT

The output of Section 4 is a uniform LSI for the formal *geometric* heat flow on a closed slice of the orbit space, with constant independent of any lattice approximation. To convert this to a mass gap for the physical Yang–Mills measure $\mu_{\text{YM}} = Z^{-1} \exp(-S_{\text{YM}}/g^2) \mathcal{D}A$ (with the appropriate regularisation), we must descend through the lattice approximations and demonstrate that the LSI constant survives the limit.

5.1. Lattice approximation. Let $\Lambda_a \subset (a\mathbb{Z})^4$ be a finite lattice of spacing $a > 0$ and side L_a . The Wilson measure $\mu_a = Z_a^{-1} \exp(-S_{\text{W},a}[U]/g^2) dU$ with $S_{\text{W},a}[U] = \sum_p (1 - \frac{1}{N} \text{ReTr} U_p)$ defines a $G^{\#\text{links}}$ -valued probability measure. Heat-bath / Langevin dynamics on μ_a are well-studied (Parisi–Wu stochastic quantisation; Brownian motion on $G^{\#\text{links}}$).

5.2. The BBD uniform LSI input. The principal external input of this Letter, beyond Theorem 3.3 and the Bakry–Émery step, is the following external assumption, modelled on the scalar φ^4 -result [7].

Hypothesis 5.1 (Uniform LSI for 4D pure $SU(N)$ Wilson measure). There exists a constant $c_\infty(D) > 0$, depending only on dimension $D = 4$ and on $G = SU(N)$, such that for every $a > 0$ and every L_a in the appropriate scaling window, the Wilson measure μ_a on the periodic Λ_a satisfies a logarithmic Sobolev inequality with constant $\rho_a \geq 1/c_\infty(D)$. Equivalently, the spectral gap of the heat-bath / Langevin generator on μ_a is at least $1/c_\infty(D)$, uniformly in a .

Status of Hypothesis 5.1. For scalar φ^4 -measures in dimensions $d = 2, 3$ this is exactly the result of Bauerschmidt–Dagallier [7] via the Polchinski equation method of [6]. For 4D $SU(N)$ pure gauge it is *open*; its derivation is the standard cluster-expansion problem in the modern probabilistic formulation. The Cao–Park–Sheffield [9] and Cao–Nissim–Sheffield [8] works on lattice Yang–Mills as random surfaces, together with the Chandra–Chevyrev–Hairer–Shen 3D construction [10], provide important recent progress but do not yet yield Hypothesis 5.1 in 4D.

5.3. Dirichlet form descent. Granted Hypothesis 5.1, the standard Fukushima–Oshima–Takeda machinery [13] for Dirichlet forms ensures that the family (μ_a) is tight in the appropriate distributional space, and that the limit point μ_∞ inherits the LSI with constant $1/c_\infty(D)$ (by Mosco-type lower semicontinuity of LSI under regular graph convergence; see [4, Ch. 4]).

5.4. Coupling KR–FP–A to Hypothesis 5.1. Two LSIs are now in play: the symbolic Bakry–Émery LSI from Proposition 4.2 with constant $(1 - \kappa_{\text{FP}})m_0^2$ on $\bar{\Lambda}_{S_0}$, and the BBD LSI with constant $1/c_\infty(D)$ on the physical Wilson measure. The coupling rests on the heuristic identification, correct in the formal continuum limit, between the symbolic heat flow on $\mathcal{A}/\mathcal{G} \upharpoonright_{\bar{\Lambda}_{S_0}}$ and the gauge-invariant component of the Wilson measure restricted to the analogous Coulomb-gauge slice. We *conjecture* (and the Bauerschmidt collaboration roadmap of Section 7 aims to formalise) that the effective LSI constant on the gauge-fixed Wilson measure is

$$\rho_{\text{eff}} \geq \frac{(1 - \kappa_{\text{FP}})m_0^2}{c_\infty(D)},$$

i.e. the product of the symbolic LSI with the BBD LSI *divided by* $c_\infty(D)$ (the standard cluster expansion contribution). This is the input that produces (2).

Remark 5.2 (Multiplicative structure of the constant). The factor $1/c_\infty(D)$ arises because the BBD LSI gives the lattice μ_a a normalised spectral gap $1/c_\infty(D)$, and the geometric component $(1 - \kappa_{\text{FP}})m_0^2$ from Proposition 4.2 *multiplies* this in the saturated Bakry–Émery scheme of [21, Sec. IV.B], where the explicit cross-group formula $C_{\text{LSI}}(G, D) = c_\infty(D) \cdot f(\pi_1(G)) \cdot [1 - \kappa_{\text{FP}} \delta_{\text{sat}}]$ has been derived empirically and shown to match $\text{SU}(N)$ lattice gauge data with mean residual 5–8% across $N = 2, 3, 4, 5$ in $D = 3, 4, 5, 6$ [21, Tab. III]. The empirical $c_\infty(4) = 0.247$ from Wilson-flow plateau measurements ([21, Sec. V]) gives, in the $\text{SU}(3)$ case, $\Delta_{\text{est}} \geq (5/6) \cdot m_0^2/0.247 \approx 3.38 m_0^2$.

6. MAIN THEOREM (CONDITIONAL)

We now state the conditional reduction in its rigorous form.

Theorem 6.1 (KR–FP–B, mass gap conditional reduction). *Let G be a compact simple Lie group with Lie algebra $\mathfrak{g}$ and positive roots Φ^+ . Set $\kappa_{\text{FP}} := 1/(2|\Phi^+|)$. Assume:*

- (i) *Hypotheses (H1), (H2), (H3) of Theorem 3.3 [20], i.e. the Faddeev–Popov spectral bound $\lambda_{\min}(d_A^\dagger d_A) \geq m_0^2(1 - \kappa_{\text{FP}})$ on $\bar{\Lambda}_{S_0} \subset \bar{\Lambda} \subset \mathcal{A}/\mathcal{G}$;*
- (ii) *Hypothesis 5.1: a uniform logarithmic Sobolev inequality with constant $1/c_\infty(D)$ for the $4D$ pure $\text{SU}(N)$ Wilson measure on the periodic lattice Λ_a , uniformly in $a > 0$ across the scaling window;*
- (iii) *Compatibility (C) of the symbolic and BBD LSIs in the sense that the gauge-fixed projection of the Wilson measure on $\bar{\Lambda}_{S_0}$ inherits both LSIs, and the resulting effective LSI constant on the orbit-space measure is at least $(1 - \kappa_{\text{FP}})m_0^2/c_\infty(D)$ in the continuum limit.*

Then the spectral gap Δ of the $4D$ pure $\text{SU}(N)$ Yang–Mills Hamiltonian satisfies

$$(4) \quad \Delta \geq \frac{(1 - \kappa_{\text{FP}})m_0^2}{c_\infty(D)} > 0.$$

In particular for $G = \text{SU}(3)$, $D = 4$,

$$\Delta \geq \frac{5}{6} \cdot \frac{m_0^2}{c_\infty(4)} > 0.$$

Proof. Step 1: By (i) and Theorem 3.3, the Faddeev–Popov spectrum on $\bar{\Lambda}_{S_0}$ is bounded below by $(1 - \kappa_{\text{FP}})m_0^2$.

Step 2: By Corollary 3.4 and the Babelon–Viallet O’Neill formula (3), $\text{Ric}_{\mathcal{A}/\mathcal{G}} \geq (1 - \kappa_{\text{FP}})m_0^2 \cdot g$ on $\bar{\Lambda}_{S_0}$.

Step 3: By Theorem 4.1 (Bakry–Émery), the symbolic Markov semigroup on $(\bar{\Lambda}_{S_0}, g_{\mathcal{A}/\mathcal{G}})$ satisfies $\text{CD}((1 - \kappa_{\text{FP}})m_0^2, \infty)$.

Step 4: By Proposition 4.4 (Otto–Villani) and Proposition 4.2, the heat-kernel measure on $\bar{\Lambda}_{S_0}$ obeys a logarithmic Sobolev inequality with constant $\rho \geq (1 - \kappa_{\text{FP}})m_0^2$.

Step 5: By Rothaus [22], the symbolic LSI implies a spectral gap $\lambda_1 \geq (1 - \kappa_{\text{FP}})m_0^2$ for the symbolic Dirichlet form on $\bar{\Lambda}_{S_0}$.

Step 6: By (ii), Hypothesis 5.1, the lattice Wilson measure μ_a has uniform LSI with constant $1/c_\infty(D)$, and the Fukushima–Oshima–Takeda Dirichlet-form Mosco-convergence machinery [13] delivers the continuum measure μ_∞ with the same LSI constant.

Step 7: By (iii), the compatibility hypothesis, the effective LSI constant on the orbit-space continuum measure $\mu_\infty^{A/G}$ is at least $(1 - \kappa_{\text{FP}})m_0^2/c_\infty(D)$.

Step 8: By the Reed–Simon transfer between LSI and the spectral gap of the associated Markov semigroup (an immediate consequence of Rothaus [22] applied to $\mu_\infty^{A/G}$), the spectral gap of the physical Yang–Mills Hamiltonian satisfies $\Delta \geq (1 - \kappa_{\text{FP}})m_0^2/c_\infty(D) > 0$.

For $G = SU(3)$: $\kappa_{\text{FP}} = 1/6$, $1 - \kappa_{\text{FP}} = 5/6$. $\square$

Remark 6.2 (The compatibility hypothesis (C) is not automatic). Compatibility (C) is the meeting point of the geometric and probabilistic analyses. It encodes the assertion that the projection of the lattice Wilson measure onto a Coulomb-gauge slice of the orbit space inherits *both* the geometric LSI (with the constant from Proposition 4.2) and the BBD LSI (with the constant from Hypothesis 5.1). Empirically (cf. [21, Tab. III]) the resulting LSI constant *multiplies* the two factors, suggesting that the Cartan-saturation correction from KR–FP–3 acts as a structural multiplicative shift on the BBD baseline $c_\infty(D)$. A rigorous proof of (C) would require a gauge-fixing argument in the BBD formalism, which we leave to future work.

Remark 6.3 (Numerical illustration for $G = SU(3)$, $D = 4$). With the empirical Wilson-flow plateau value $c_\infty^{\text{emp}}(4) = 0.247 \pm 0.001$ ([21, Tab. II]), the conditional bound reads

$$\Delta \geq \frac{5}{6} \cdot \frac{m_0^2}{0.247} \approx 3.38 m_0^2.$$

With $m_0^2 \sim \Lambda_{\text{QCD}}^2 \sim (250 \text{ MeV})^2$ this gives $\Delta \gtrsim (460 \text{ MeV})^2$, of the correct order of magnitude as the lattice glueball mass $m(0^{++})_{SU(3)} \simeq 1700 \text{ MeV}$ (Morningstar–Peardon [18]). The agreement should not be overstated: the absolute prefactor m_0^2 is unknown *a priori*, and only the dependence on κ_{FP} and $c_\infty(D)$ is delivered by Theorem 6.1.

7. HONEST CONDITIONALITY AND FORMALISATION ROADMAP

We list explicitly the remaining open problems and provide a tentative roadmap.

7.1. Open hypotheses.

- **(H1) Generic-vanishing** [20]. The asymptotic alignment of minimising eigenfunctions $\eta_{\min}(A)$ of $M[A]$ with the Cartan subspace $\mathfrak{h} \otimes \Omega_{\text{transverse}}^1$ near $\partial\Omega$. Numerical evidence in $SU(2)$ lattice computations [20, Table 1 ($SU(2)$ lattice)] (Caveat 1: matching to $\kappa_{\text{FP}} = 1/6$ rather than $\kappa_{\text{FP}}^{SU(2)} = 1/2$ requires further analysis). Theoretical proof requires perturbation analysis on $\bar{\Lambda}_{S_0}$ near the horizon; estimated effort 4–7 months full-time.
- **(H2) Compact-manifold Sobolev constant**. The optimal Aubin–Talenti constant $C_S = (3/(4\pi^2))^{1/4}$ is for the unbounded $\mathbb{R}^4$. On the compact M a quantitative lower-order correction is needed, depending on $\text{vol}(M)$, $\text{inj}(M)$, $\text{Ric}(M)$. This is a standard but unwritten estimate. Effort: 1–2 months.
- **(H3) Measurable Cartan selection on $\bar{\Lambda}$** . Standard at regular points; care required at degeneracy strata. Hahn–Banach plus measurable selection theorems should suffice. Effort: 1 month.
- **Hypothesis 5.1 (BBD uniform LSI for 4D pure $SU(N)$)**. The principal external input. The scalar version is in [7]; the non-Abelian gauge adaptation involves substantial technical work (Polchinski equation for $SU(N)$ Haar measure, cluster expansion control). Estimated effort with Bauerschmidt collaboration: 18–24 months. A first email contact has been initiated by the author (May 2026).
- **Compatibility (C)**. Coupling of geometric and BBD LSIs via a gauge-fixed projection. Conditional on the gauge-fixing being implemented in the Polchinski-RG framework. Effort: 2–4 months conditional on the BBD-collaboration framework.

7.2. Path probability estimate. Following the convention of [21, Tab. IV], we quantify the probability that the full chain closes within 10 years as follows: under the Bauerschmidt-collaboration path, conditional on (H1)–(H3) and Hypothesis 5.1 all being discharged, $P(\text{Theorem 6.1 formalised within } 10\text{y}) \approx 45\text{--}60\%$, i.e. the dominant uncertainty is on Hypothesis 5.1 for $SU(N)$ and on the compatibility (C). Conditional on the direct path (H1)–(H3) only (no collaboration), $P \approx 35\text{--}50\%$ at the price of 4–7 months on (H1) alone.

7.3. Strict bypass note. The reduction articulated here does *not* bypass the Balaban–Bauerschmidt cluster-expansion programme. It bypasses the cluster expansion *for the Faddeev–Popov spectral lower bound* (Theorem 3.3 replaces it with a geometric/Lie-algebraic argument), but still requires the cluster expansion (in the BBD form, Hypothesis 5.1) for the measure descent. This is a partial bypass.

8. CONCLUSION

We have articulated a conditional reduction of the 4D pure $SU(N)$ Yang–Mills mass gap problem along the chain

$$\text{KR–FP–3} \Rightarrow \text{KR–FP–A (Babelon–Viallet O’Neill)} \Rightarrow \text{Bakry–Émery} \Rightarrow \text{Otto–Villani LSI} \Rightarrow \text{Rothaus gap}$$

The output is the bound $\Delta \geq (1 - \kappa_{\text{FP}}) m_0^2 / c_\infty(D)$, which for $G = SU(3)$ in $D = 4$ reads $\Delta \geq \frac{5}{6} m_0^2 / c_\infty(4)$. The contribution is structural: a single conceptual chain that exhibits explicitly which open analytic problems remain. These are the three KR–FP–3 hypotheses (H1)–(H3) of [20], the BBD-style uniform LSI Hypothesis 5.1 for 4D pure $SU(N)$ gauge in the spirit of [6, 7], and the compatibility (C). We do *not* claim a proof of the Clay mass gap.

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